Prosper Magara – Curriculum Vitae

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Contact

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Available to relocate to KwaZulu-Natal and to travel up to 50% of the time.

Professional summary

Biomedical and electrical engineer with medical equipment field experience since 2020 and a PhD focus on laparoscopic surgery. I have installed, calibrated and serviced X-ray, dialysis, autoclave and laboratory equipment in hospitals across Zimbabwe, and now work with surgeons at Groote Schuur Hospital as a Google PhD Fellow at the University of Cape Town on telementoring for laparoscopic surgery, with occasional work on da Vinci robot-assisted cases. Methodical troubleshooter, careful record-keeper and a clear communicator with clinical teams.

Core competencies

- **Field service:** installation, commissioning, calibration, preventive maintenance and repair of X-ray, dialysis, autoclave, theatre and laboratory equipment; spares and service reports.
- **Laparoscopy and surgical robotics:** laparoscopic video chains and instruments; hands-on training on the LapSim simulator and box trainers; exposure to the da Vinci surgical robot in theatre; operating-theatre protocol.
- **Diagnostics and software:** electro-mechanical and embedded troubleshooting (PIC, STM32, ATmega328); remote diagnostics and latency tooling; Windows and Linux; Python, C#, JavaScript; Wi-Fi, GSM, 5G, WebRTC.
- **People and admin:** working relationships with surgeons and hospital staff; service records and inventories; technical writing; supervising junior engineers.

Professional experience

PhD Researcher and Google PhD Fellow, Division of Biomedical Engineering, University of Cape Town · 2024 to present

- Telementoring for laparoscopic surgery with surgical teams at Groote Schuur Hospital, with occasional robot-assisted cases; exposure to the da Vinci system in theatre and hands-on training on the LapSim simulator and box trainers.
- Built a real-time telementoring platform (dual-camera WebRTC streaming, telestration, two-way audio) and a glass-to-glass latency rig (ATmega328 microcontroller, light sensors, desktop software) for diagnosing video delay.
- Lead author of a systematic review of AI bleeding monitoring in laparoscopic and robot-assisted surgery (Journal of Robotic Surgery, 2026) and a latency-aware evaluation of bleeding detection algorithms (Springer, 2026).

Medical Device Engineer, Med-Elect Technologies, Zimbabwe · 2022 to 2023

- Planned and carried out installations of X-ray machines, dialysis machines, autoclaves, theatre beds and chemistry analysers at hospitals and clinics.
- Serviced, calibrated and repaired equipment on site; scheduled preventive maintenance, kept service records and handled handover and user training with clinical staff.

Biomedical Engineering Attaché, Parirenyatwa Group of Hospitals, Harare · 2022

- Equipment inventory, installation and maintenance across surgery, paediatrics, oncology, haematology, orthopaedics and radiography; took part in a new MRI installation and a water-treatment plant installation.

Biomedical and IoT Engineer, SolidarMed, Zimbabwe · 2020 to 2021

- Developed IoT health devices, implemented telemedicine tooling (REDCap), wrote technical reports and supervised four students on attachment.

Freelance Solar Systems Engineer, Zimbabwe · 2018 to 2023

- Designed, installed and maintained off-grid and hybrid solar PV systems for homes, farms and clinics.

Education

- **PhD, Biomedical Engineering**, University of Cape Town, 2024 to present. Telementoring and telesurgery with augmented reality and haptics over 5G.
- **MSc, Biomedical Engineering**, University of Zimbabwe, 2021 to 2022, distinction. Thesis: a smart health station for swab collection and patient data entry.
- **BSc (Hons), Electrical Engineering**, University of Zimbabwe, 2015 to 2019, upper second class.

Awards and fellowships

- 2026: PhD First Prize, Best Presenter Award, BME Ignite (HUB Biomedical Engineering Fusion event), University of Cape Town.
- 2024: Google PhD Fellowship, for research combining AI and augmented reality in telementoring for laparoscopic surgery.
- 2019: Best Innovative Project, University of Zimbabwe Research Week, for the smart egg incubator.
- Second Creative SDGs Idea, CBZ Bank Hacking for SDGs post COVID-19 hackathon, Zimbabwe.

Publications

1. Magara P, Sivarasu S, Ras L, Biyabani A, Blocker A, Shabangu M, Malila B. AI-based automated bleeding monitoring in conventional and robot-assisted laparoscopic surgery: a systematic review. *J Robot Surg* 2026;20(1):522. Link
2. Magara P, Sivarasu S, Ras L, Biyabani A, Malila B. Latency-aware evaluation of bleeding monitoring algorithms for real-time surgical applications. *Nordic Conference on Digital Health and Wireless Solutions*, Springer, 2026, pp. 51–69. Link
3. Blocker A, Magara P, Biyabani A, Datay MI, Mwangama J, Malila B. Design of a solar-based virtual clinic for austere environments. *IEEE AFRICON* 2025, pp. 1–8. Link
4. Magara P, Ras L, Biyabani A, Malila B. User-centred design of a tele-mentoring system for laparoscopic surgery in austere environments. *IEEE Globecom Workshops* 2024, pp. 1–7. Link
5. Magara P, Saruchera G, Mushiri T, Noorbhai H. Development of a semi-automatic, isolating and portable health-station for swab collection and patient data entry. University of Zimbabwe, 2022.

Additional information

- Languages: English and Shona, both native.
- Driver's licence: valid licence with a clean driving record.
- Research groups: Telemedicine and Connected Care (2024 to 2027) and UCT MedTech (2025 to 2027), University of Cape Town.
- Research travel: Rwanda, Italy and Finland.
- Referees: available on request.